

DATA STRUCTURE ALGORITHM-II

SORTING: Sorting is the process of arranging data in a particular order, usually in ascending or descending order. Sorting makes the operations faster and easier.

Eg: Searching

Types of Sorting Algorithms				
Sorting Algorithm	Best	Average	Worst	Space
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$

BUBBLE SORT → Repeatedly compare adjacent elements and swap them if they are in the wrong order.

SELECTION SORT → Find the smallest element and place it at the correct position.

INSERTION SORT → Insert each element into its correct position in the already sorted part.

MERGE SORT → Divide the array into halves, sort each half recursively, then merge them.

QUICK SORT → Choose a pivot, partition the array, then recursively sort the left and right parts.

NOTE:

One-Page Revision Table		
Scenario	Best Choice	Reason
Nearly sorted	Insertion Sort	Best case $O(n)$
Small array	Insertion Sort	Low overhead
Guaranteed $O(n \log n)$	Merge Sort	Worst case $O(n \log n)$
Fastest average	Quick Sort	Excellent practical performance
Stable sort required	Merge Sort	Stable
Memory limited	Quick Sort	In-place
Minimum swaps	Selection Sort	Only $n-1$ swaps
Basic learning	Bubble Sort	Easy to understand
Linked List	Merge Sort	Efficient for linked lists

HASH TABLE: A Hash Table is a non-linear data structure that stores data as key-value pairs and provides fast insertion, deletion, and searching using a hash function.

A hash table is used when we need fast insertion, deletion and searching using hash function.

Q: What is hashing: It is the process of converting a key into an index using a hash function

A hash function is a mathematical function that converts key into an index of the hash table.

Q: What is collision and how collision occurs: A collision occurs when two or more keys produce the same hash index.

There are 2 types of collision resolution:

- A) Separate Chaining (Open Hashing) → It stores all colliding index in a list at the same index.
- B) Open Addressing (Closed Hashing) → Instead of storing multiple elements at one index, OA finds another empty position in the table.

Types of open addressing

1. Linear probing: Searches next available slot after a collision.
2. Quadratic probing: Searches using quadratic increments.
3. Double hashing: Use a second hash function to determine next location.

STACK AND QUEUE: A Stack is a linear data structure that follows the LIFO (Last In, First Out) principle.

Applications: Undo/Redo in editors, DFS

A Queue is a linear data structure that follows the FIFO (First In, First Out) principle.

Applications: Printer queue, BFS

TREE: A Tree is a non-linear hierarchical data structure consisting of nodes connected by edges. It has one root node, and every node can have zero or more child nodes.

Trees are generally used when data naturally forms a parent child relationship or when we need fast insertion, searching and deletion.

Application: BST, Database Indexing

Binary Tree is a tree in which each node has at most two children

Binary Search Tree (BST): Is a BT in which the left subtree contains values smaller than the root and right subtree contains greater than the root.

BST allows fast insertion, deletion and searching because we eliminate half of the remaining tree at each step.

GRAPH: A Graph is a non-linear data structure that consists of a set of vertices (nodes) and edges connecting those vertices.

Graphs are used to represent relationships or connection between objects.

Application of graphs: Google map, social networks

types of graphs: Undirected (Edges have no direction), directed graph (Edges have directions) and weighted graph (Edges have weight)

Graph Traversal → It is the process of visiting every vertex (node) of a graph exactly once in a systematic way.

Two types of graph traversal → BFS (Breadth First Search): It visits all neighboring nodes first before moving to the next level.

DFS (Depth First Search) → DFS explores one path completely before backtracking and exploring another path.